

# Direct synthesis of single-layer graphene films on quartz substrate by a nanoparticle-assisted method

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## ARTICLE INFO

### Keywords:

Graphene  
Quartz substrate  
CVD  
Nanoparticle  
Defogge

## ABSTRACT

Matching graphene to semiconducting or insulating substrates is necessary for wide-ranging applications. Although synthesis of high-quality graphene on metal surfaces has gained great success, the transfer process of graphene is quite complicated and may cause defects and contamination. Therefore, the growth of graphene on insulating substrates is very important. Here, a direct synthesis method of single-layer graphene films on quartz substrate via a Cu-nanoparticle-assisted method was reported (N-CVD). The growth rate and quality of graphene on quartz showed a remarkable improvement compared with the conventional CVD process. Localized catalysis made it possible for large-scale and continuous growth. Furthermore, the graphene glass fabricated by this method showed great performances on electrical, optical and thermal properties, and was proved to be hydrophobic. We designed a 2 cm × 2 cm single-layer graphene defogger, which can remove the fog completely within 10 s with a 30 V input voltage. Briefly, this work aims to present a low-cost, uniform and scalable method for graphene synthesis on insulating substrates.

## 1. Introduction

Graphene, the first two-dimensional material with carbon atoms linked in a honeycomb pattern, has received extensive attention due to its superior properties [1–3]. In particular, extraordinary mechanical strength, excellent electrical/thermal conductivity, and high optical transparency make it suitable for transparent conducting electrodes [4], photodetectors [5,6], supercapacitors [7], and photocatalysis devices [8]. Being compatible with semiconducting or insulating substrates is a necessary condition for achieving the above potential applications. However, the transfer process of graphene between metal substrate and device-compatible substrate is complex, which is a major issue for large-scale and industrial production [9]. The quality of graphene may also degrade due to defects and contamination caused by transferring (wrinkles, holes, and cracks) process, which can result in limitations of device performance [10]. Therefore, the direct synthesis of high-quality and large-area graphene films on insulating substrates is very intriguing [11,12].

Chemical vapor deposition (CVD) has emerged as the most promising method for industrial production due to its excellent repeatability to synthesis high-quality and large-area graphene [13,14]. In recent years, direct graphene growth on insulating substrates, such as SiO<sub>2</sub> [15,16], hexagonal boron nitride [17] and Si<sub>3</sub>N<sub>4</sub> [18], has been

successfully realized by a CVD process. However, the extremely low catalytic activity of such substrates resulted in the lack of pyrolysis of carbon precursors and the low mobility of carbon atoms, which led to a very long growth time. Introducing catalytic metal materials into the conventional CVD process was proved to be an effective way to reduce the growth time and improve the graphene quality [19]. Some strategies have been attempted by using the metal vapor as a catalyst, resulting in a notable improvement in the quality and growth rate of graphene [20,21]. However, it is hard to guarantee the uniformity of metal vapor when synthesis a large-scale sample, which may cause the uneven of graphene film. Other efforts, such as interfacial segregation growth or sacrificial metal coating approach, have been proved to be useful ways to increase the growth rate as well [11,22,23], but included a rather complex process before CVD growth and the sacrifice of metal is high-cost. Thus, a low-cost, scalable and uniform catalyst method for graphene growth is highly demanded.

Metal nanoparticles have excellent properties compared with normal metals, especially in the field of catalysis [24]. On the other hand, industrial processes with low cost have been developed for many metal nanoparticles, such as Cu, Ni and Cr. Thus, such particles are very suitable for industrial catalysis. In previous studies, Cu nanoparticles have shown excellent performances for graphene nucleation during CVD growth [25], and have been widely used in the fabrication of

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<https://doi.org/10.1016/j.apsusc.2020.147082>

Received 28 January 2020; Received in revised form 24 June 2020; Accepted 24 June 2020

Available online 30 June 2020

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carbon nanotube [26]. The above indicates that graphene growth on insulating substrates may also succeed with the assist of Cu nanoparticles.

In this work, we designed a Cu-nanoparticle-assisted method to realize the direct growth of high-quality and large-area graphene films on quartz substrates, which was named as N-CVD. Cu nanoparticles were introduced on quartz by uniform dispersion and spin coating. During the N-CVD growth process, each particle is responsible for the catalysis of a micrometer scale area, which is able to lower the barrier of methane cracking and increase the rate of graphene growth. We further fabricated a defogger with graphene glass, which showed an excellent performance in defogging rate.

## 2. Experimental section

### 2.1. Preset of Cu nanoparticles

Prior to the synthesis of graphene, quartz substrates were cleaned ultrasonically (15 min for each liquid) and rinsed with deionized water, acetone and alcohol. Cu nanoparticles (Shanghai Cw-Nano Co., Ltd, average particle size is 20–30 nm) were evenly dispersed into alcohol with a concentration of 5.00 mg·ml<sup>-1</sup> using ultrasonic method. Then, the dispersion was spin-coated on the as-prepared quartz substrate and placed on a heating plate (~40 °C) for 5 min to obtain a quartz substrate with uniform preset of “catalytic seeds”.

### 2.2. Growth of graphene

Graphene films are grown on quartz substrate by Atmospheric Pressure CVD (APCVD) with the assist of Cu nanoparticles. Quartz substrate with “seeds” was placed in a commercially available tube furnace (Xiamen G-CVD Material Technology Co., Ltd). The tube was pumped below 0.5 Pa to remove air, and then stopped pumping. The temperature was heated to the growth temperature (1100 °C) with 360 sccm Ar and 96 sccm H<sub>2</sub>, the pressure inside the tube gradually approaches atmospheric pressure. After the temperature stabilized (15 min), the Ar/H<sub>2</sub> gas were converted to 120 sccm and 22 sccm respectively, and CH<sub>4</sub> (1.5 sccm) gas was fed into the reactor to initiate the graphene growth on quartz surface. After the growth, the sample was post-treated under Ar/H<sub>2</sub> for 20 min and then cooled down by sliding the heated zone away.

### 2.3. Characterization of graphene

The prepared samples were characterized sequentially by optical microscopy (Olympus BX51), Raman spectroscopy (Thermo DXRxi, with laser excitation at 532 nm, and power of 7mW), SEM (Hitachi S-4800, operating at 1 kV), UV-vis spectroscopy (Perkin-Elmer Lambda 950 spectrophotometer, detecting at 300 nm–800 nm), and four-probe resistance tester (CDE, Res Map 178), contact angle tester (Biolin, THETA).

### 2.4. Fabrication and measurement of graphene defoggers

Cu wires were pasted on both edges of graphene glass with conductive silver glue. The input voltages were applied to the defoggers by DC power supply (HYELEC HY3005ET). The deionized water was heated to ca. 67.3 °C to produce steam continuously. Temperature of both graphene glass and deionized water was measured with K-type thermocouples and digital thermometer (SMART SENSOR AS877).

## 3. Results and discussion

Fig. 1a schematically illustrates the seed assisted catalytic process, which uses Cu nanoparticles to decompose the methane feedstock. In order to achieve this, Cu nanoparticles were dispersed into alcohol and

spin-coated on a piece of quartz (2 × 2 cm<sup>2</sup>). The distribution method can also be replaced by blade coating when synthesis a bigger sample (2 × 6 cm<sup>2</sup>). As shown in Fig. S1, the average distribution of seeds can be precisely controlled (1 × 10<sup>5</sup> ~ 2 × 10<sup>6</sup> particles cm<sup>-2</sup>), which has an effect on both graphene quality and transmittance of graphene glass. Heating after distribution helps the alcohol volatilize faster and reduce clustering of nanoparticles. Fig. 1b schematically illustrates the seed-assisted catalytic process. Cu nanoparticles are like signal towers, and each seed is responsible for a small area, which resulted in the formation of larger and more uniform graphene films.

Raman spectroscopy is a powerful technique for characterizing the structure and electronic properties of graphene [27,28]. Fig. 2 shows typical Raman spectra of graphene on quartz with and without the assist of Cu nanoparticles. As shown in Fig. 2a, the G peak (~1580 cm<sup>-1</sup>) associated with the planar configuration sp<sup>2</sup> bonded carbon begins to appear at the growth time of 30 min, indicating the nucleation of graphene with the assist of Cu nanoparticles. The D peaks (~1350 cm<sup>-1</sup>) represent the breathing modes from carbon rings and can be active near the graphene edges or defects [29,30]. As the growth time increase, the D peak decreases gradually, suggesting a significant decrease in graphene edges and defects during the growth of graphene.

The 2D peak (~2700 cm<sup>-1</sup>) is due to a two-phonon lattice vibrational process and is referred to as an overtone of the D peak. It does not represent defects and is often used to determine the quality and thickness of graphene. Notably, for the longest growth time (240 min), the intensity ratio of the 2D peak and the G peak is larger than 1.4; the ratio of the integral area's values of 2D and G peaks is larger than 3.0; Moreover, the 2D peak has a symmetric shape with a single Lorentzian profile, indicating that monolayer graphene film was predominantly prepared. More details could be found in the Supporting Information (S5). To investigate the role of Cu nanoparticles in graphene growth, we designed a comparison experiment (conventional CVD method without Cu-nanoparticle-assisted) to show the differences in growth behavior between the seed-assisted growth and the non-seed-assisted growth. It is worth noting that pristine and “seeded” samples were placed into the furnace at the same time under each condition. Fig. 2b illustrates the Raman shift of the control group. The growth rate of graphene was extremely slow on pristine quartz, no peaks were observed within 60 min, and the quality of graphene was lower under each condition as well.

Signal to noise ratio (SNR) is also a parameter worth noting. As shown in Fig. 2c, SNR of the control group is much lower, although all Raman spectra presented in this work were recorded under the same instrument and test parameters. The SNR reflected the relative content of graphene. The diameters of graphene domains grown on quartz were about 100 nm to 500 nm, while the Raman spot size was larger than 1 μm. Higher quantity of graphene domains in the Raman spot leads to a higher SNR. As mentioned before, the nucleation and growth rate of graphene on pristine quartz was extremely slow compared with the “seeded” sample, which leads to smaller coverage and lower SNR. This is also confirmed in the SEM micrographs and square resistance discussed later.

Fig. 3 showed the Raman mapping images and corresponding optical images of the graphene grown on quartz substrates with or without nanoparticles (under 240 min growth). As for the “seeded” sample, more than 98% of the area has I<sub>D</sub>/I<sub>G</sub> ratios of 0.4–0.8, and more than 95% of the area has I<sub>2D</sub>/I<sub>G</sub> ratios ranging from 1.4 to 1.8, suggesting the formation of uniform and low-defect monolayer graphene. It is noteworthy that some small green spots (indicate more defects) refer to areas where nanoparticles exist (marked by the dotted line between Fig. 3a and b). These nanoparticles larger than 20 nm may originate from clustering during the dispersing and spin-coating process with little impact on uniformity of graphene for limited distribution. This problem can be regulated by tuning the concentration of the dispersion or a better dispersing solvent. In contrast, pristine sample has a non-uniform Raman mapping (Fig. 3e and f). Less than 47% of the area has

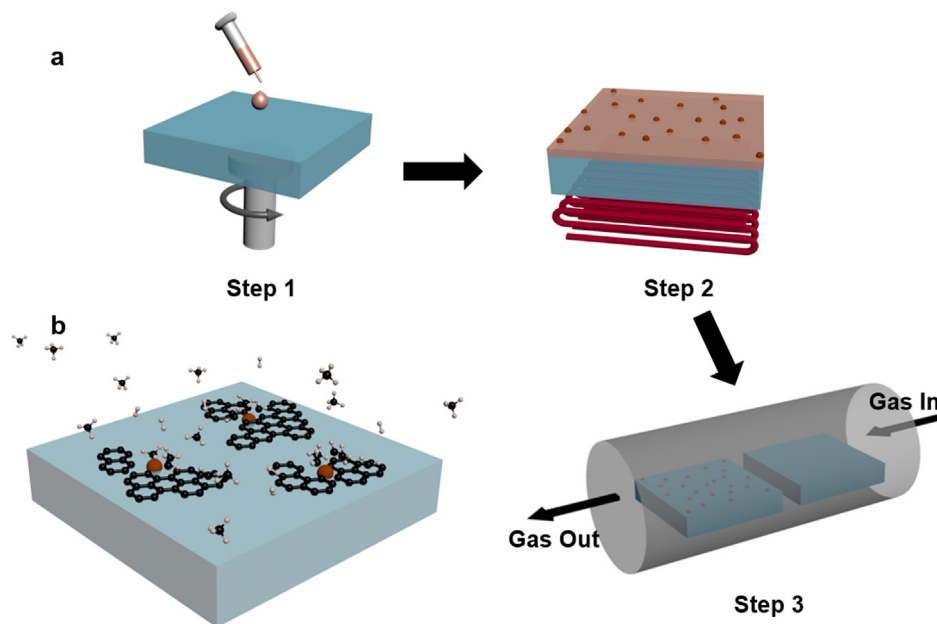


Fig. 1. (a) Experimental process of direct growth of graphene on quartz substrate via N-CVD method. Nanoparticles were spin coated on quartz followed by a short heating. The control group was placed in tube furnace at the same time during CVD growth; (b) Schematically illustration of nanoparticle-assisted catalysis.

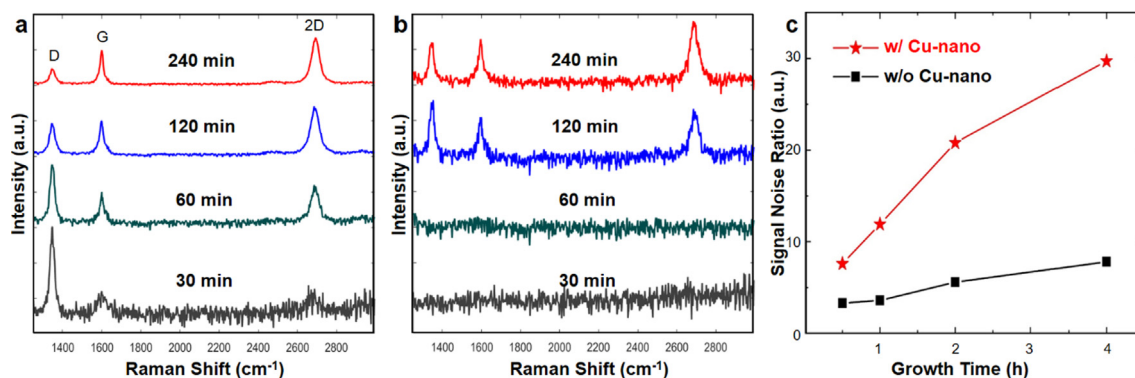


Fig. 2. (a) Raman spectra of the time-dependent growth of graphene on quartz substrate for Cu-nanoparticle-assisted method; (b) Raman spectra of the time-dependent growth of graphene on quartz substrate for control group; (c) SNR comparison chart of Raman spectra.

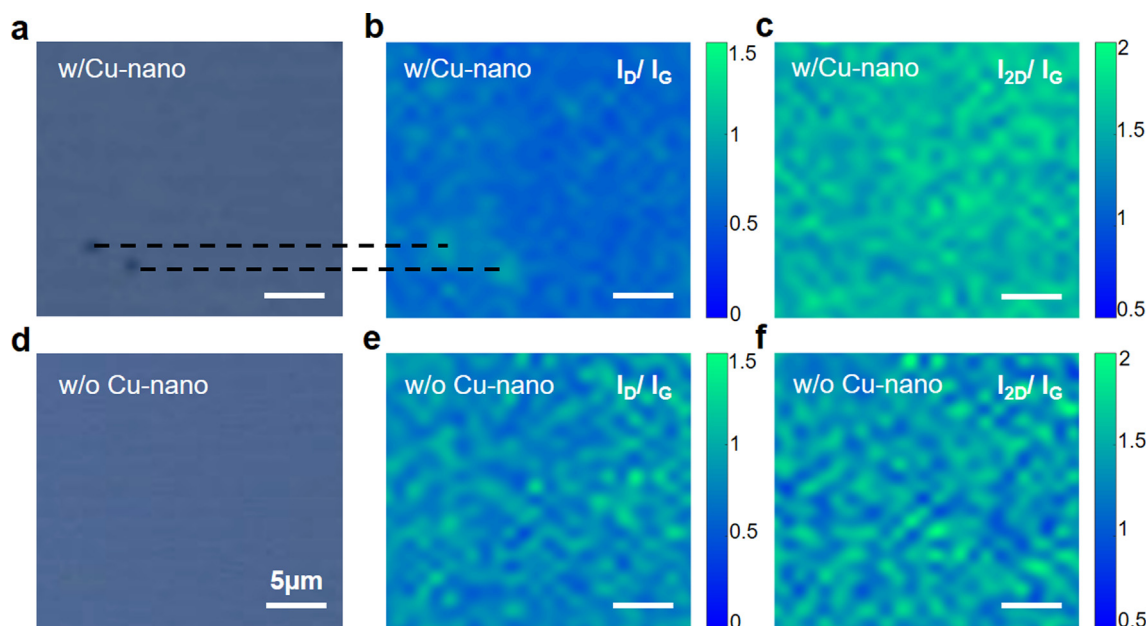
$I_D/I_G$  ratios of 0.4–0.8, 10% of them are larger than 1.0, and only 50% of the area has  $I_{2D}/I_G$  ratios ranging from 1.4 to 1.8. This indicates that a non-uniform and defective graphene glass was formed with the regular method in 240 min, longer growth time or higher temperature is needed.

Nanoparticle-assisted growth is expected to achieve the production of uniform and low-defect graphene glass of any size due to the local catalysis on the micrometer scale. In order to verify this, we pre-set Cu nanoparticles on 2 cm × 6 cm quartz by blade coating, and synthetic graphene film with the same growth process as mentioned before. The graphene glass obtained was ideal (Fig. S2). Subject to the diameter of the quartz tube and the length of the constant temperature zone (smaller than 10 cm), we did not try a larger size of quartz.

To understand the evolution of the graphene growth for the regular process and seed-assisted process, we performed scanning electron microscopy (SEM) images of the as-synthesized graphene on quartz of different growth time. The graphene area and the quartz substrate can be distinguished by the different contrasts [31]. Unlike growing on metal substrate, graphene tends to grow isotropically on quartz, graphene domains were distinguished as “round disks”. As for “seeded” sample, most “disks” were connected to each other and formed a continuous film. A single and full “disk” was pointed out in Fig. 4c. The SEM morphology of graphene directly grown on quartz was also

discussed by previous reports [19,31,32]. With the assist of Cu nanoparticles, the coverage of graphene has been estimated to be 88.48% for 120 min growth and 99.76% for 240 min growth. Cu nanoparticle and graphene domains were also marked with numbers in Fig. 4b and c. As for the pristine sample, the coverage of graphene for 120 and 240 min growth is lower than 2% and 10%, respectively. This means graphene glass needs longer growth time without the assist of Cu nanoparticles. With the increase of growth time, the density of nucleation sites and the area of graphene domains increase gradually. Due to the low mobility of carbon atoms on quartz substrate [19], it is difficult to continue to grow after the diameter of graphene domains reached about 300 nm. As is seen clearly in above figures, the nucleation density in the N-CVD process is much higher than that of the conventional CVD process, so does the area of domains. Catalytic seeds were distributed evenly on quartz and lead to larger quantity, uniform and complete of methane cracking.

To further confirm the uniformity and high electrical conductivity of the graphene films on quartz (240 min growth), we used the four-probe method to measure the sheet resistance at 25 points over a 2 × 2 cm<sup>2</sup> graphene glass (transparency ~ 94.76% at 550 nm). As shown in Fig. 5a, the sheet resistance of the graphene film grown with the assist of Cu nanoparticles (5.00 mg·ml<sup>-1</sup>) has a range of 2.71 ~ 3.26 kΩ·sq<sup>-1</sup>. The uniformity of orange color in most areas

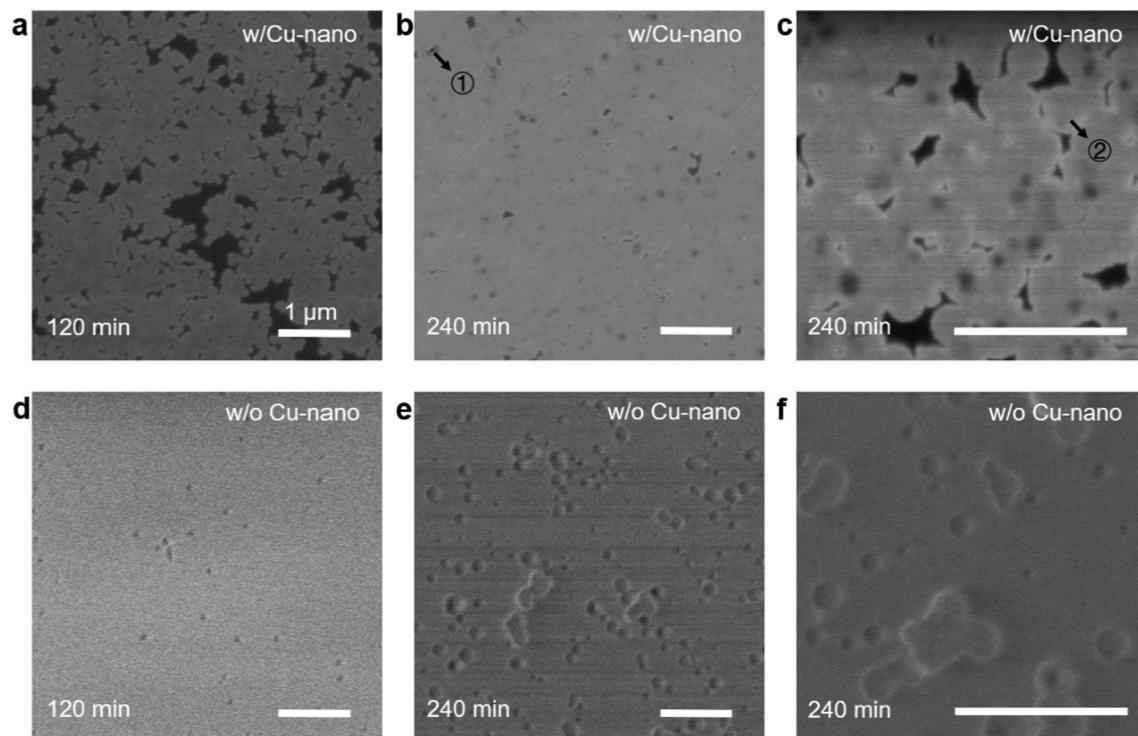


**Fig. 3.** (a–c) Optical image and corresponding Raman mapping images of the graphene glass with Cu nanoparticles. The black dashed lines between figure a and b show the defect area caused by clusters of nanoparticles; (d and e) Optical image and corresponding Raman mapping images of the control group.

illustrates the uniform distribution. As a single-layer graphene glass, the average sheet resistance was calculated as  $2.91 \text{ k}\Omega\text{sq}^{-1}$ , superior to most of the single-layer graphene glasses demonstrated by previous reports (e.g.,  $3 \text{ k}\Omega\text{sq}^{-1}$ ,  $3.2 \text{ k}\Omega\text{sq}^{-1}$ ,  $6.1 \text{ k}\Omega\text{sq}^{-1}$ ) [32–34]. This was attributed to the localized catalyzation, which leads to fewer defects and more uniform of the graphene film. Very interestingly, although there was no obvious difference in the Raman intensity ratios between the “seeded”/pristine samples (Fig. 2), the difference of sheet resistance was quite significant. Poor connectivity of graphene glass without the assist of Cu nanoparticles leads a high sheet resistance, which exceeded

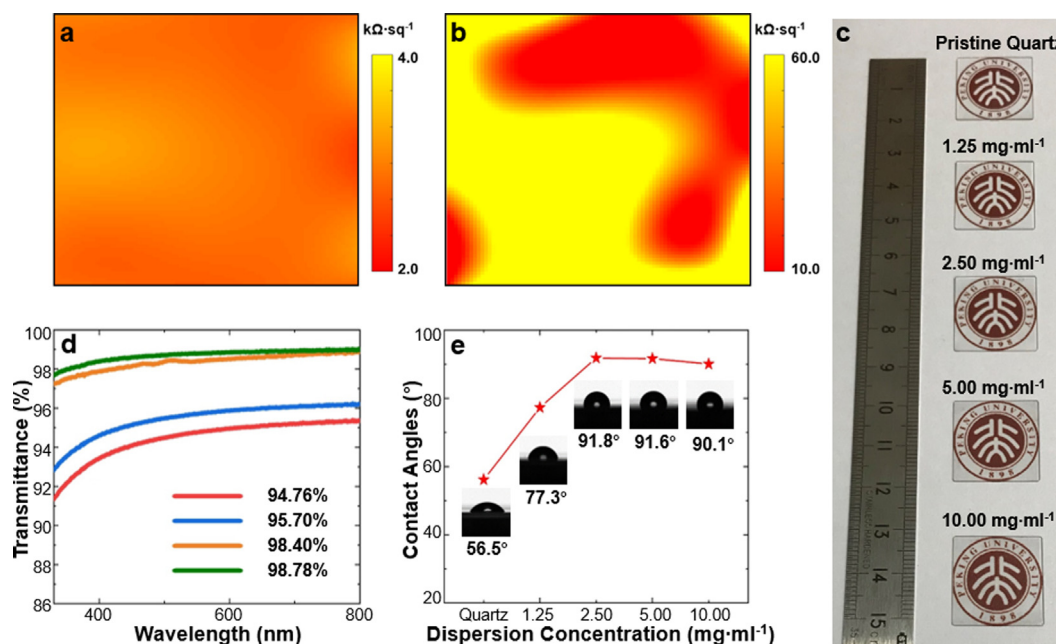
the range of the four-probe device ( $1 \text{ m}\Omega\text{sq}^{-1} \sim 5 \text{ M}\Omega\text{sq}^{-1}$ ). Pristine samples need much longer growth time to increase the coverage of graphene film in order to achieve the potential of application.

As mentioned before, the average distribution of Cu nanoparticles, which can be adjusted by the concentration of dispersion, showed an effect on the quality and transmittance of graphene glass. We altered the concentration from  $1.25 \text{ mg}\cdot\text{ml}^{-1}$  to  $10.00 \text{ mg}\cdot\text{ml}^{-1}$ , and subsequently synthesized graphene glass. For convenience, the samples were defined as sample 1 ~ 5 ( $0 \text{ mg}\cdot\text{ml}^{-1}$ ,  $1.25 \text{ mg}\cdot\text{ml}^{-1}$ ,  $2.50 \text{ mg}\cdot\text{ml}^{-1}$ ,  $5.00 \text{ mg}\cdot\text{ml}^{-1}$ ,  $10.00 \text{ mg}\cdot\text{ml}^{-1}$ , respectively). Further measurements



**Fig. 4.** SEM characterizations of graphene films on quartz substrate. The scale bar is 1 μm for all the images. (a–c) SEM images of “seeded” samples for 120 min and 240 min, ① in Figure b shows a Cu nanoparticle pointed with an arrow, ② in Figure c pointed out a graphene domain; (d–f) SEM images of pristine samples.





**Fig. 5.** (a and b) Sheet resistance mapping of “seeded” samples, for Figure a and b, dispersion concentration is  $5.00\text{ mg}\cdot\text{ml}^{-1}$  and  $1.25\text{ mg}\cdot\text{ml}^{-1}$ , respectively. 25 points of data were taken. The figures were smoothed by interpolating method; (c) Photographs of pristine quartz (the top one, without CVD process) and graphene glass; (d) UV-vis transmittance of the graphene glass. The green, yellow, blue and red lines are sample 1 ~ 4, respectively; (e) Contact angles of pristine quartz and graphene glass, the insets in (e) show the photographs of samples with a drop of water.

were presented in Fig. 5.

Sample 1 was the pristine sample, the sheet resistance of sample 1 exceeded the range of measurement. The sheet resistance contour of sample 2 was presented in order to compare the “seeded” sample (Sample 4). As shown in Fig. 5b, the data were ranged between  $9.46 \sim 53.2\text{ k}\Omega\cdot\text{sq}^{-1}$ , while 14 points exceeded the measurement range. The average sheet resistance of 11 points within range was  $16.5\text{ k}\Omega\cdot\text{sq}^{-1}$ , which was not suitable for the fabrication of devices.

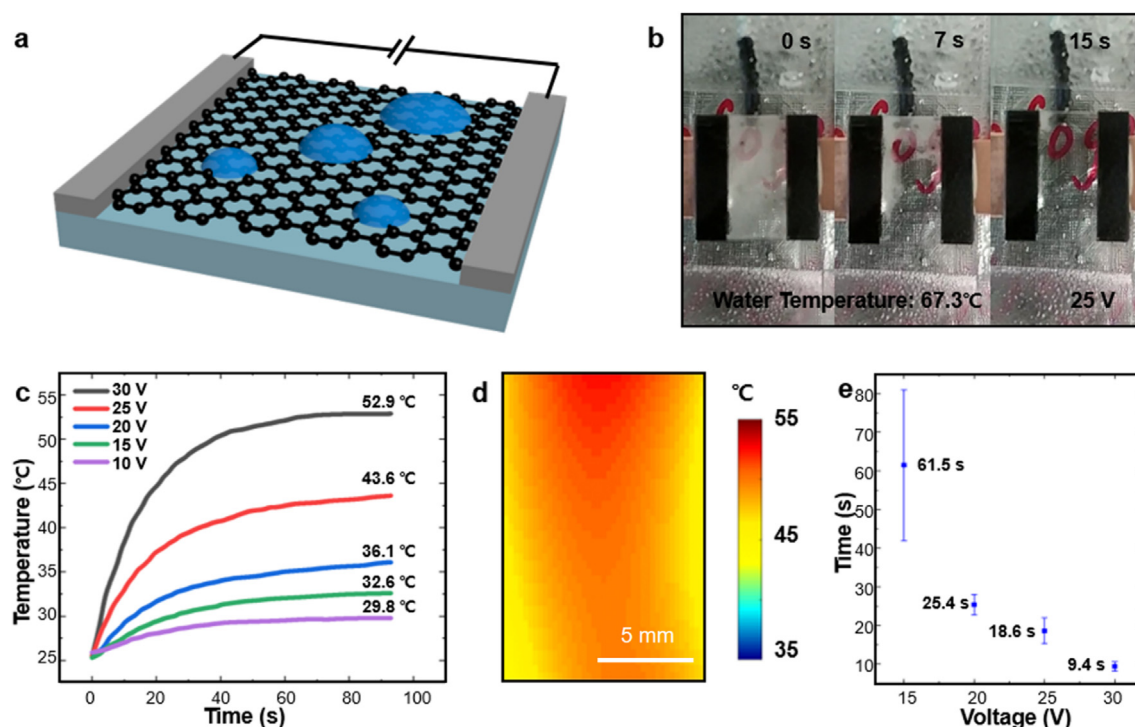
Fig. 5d is the UV-vis spectroscopy of sample 1 ~ 4, the background of air and quartz were automatically deducted. Based on the results above, our single-layer graphene glass (sample 3,  $94.76\%$  at  $550\text{ nm}$ ) had an ideal optical property, which was very close to the optimal values of mechanically exfoliated graphene ( $97.3\%$ ). It's worth noting that graphene was inevitably grown on both size of quartz (lower coverage of graphene on the back of the quartz). The transmittance of sample 5 ( $\sim 85.57\%$  at  $550\text{ nm}$ ) was far below sample 1 ~ 4 and so was not shown here.

The graphene grown on quartz can also tune the wettability of the substrates [35,36]. Fig. 5e showed the contact angle of deionized water on pristine quartz and graphene glass. The contact angle of pristine quartz is  $56.5^\circ$ , which is hydrophilic. Higher dispersion concentration led to a higher coverage of graphene, and the subsequent increase of contact angle (up to  $\sim 91^\circ$ ). The contact angle was slightly reduced when more Cu nanoparticles were applied (contact angle of Cu is  $\sim 10^\circ$  at room temperature [37]). Such performance of our graphene glass showed potential for the applications as self-cleaning windows and defoggers.

Excellent electrical, thermal, optical and hydrophobic properties made graphene glass an ideal material for transparent defogging device [38]. As shown in Fig. 6b, our single-layer graphene defogger was covered with fog before we input the voltage. A steady flow of steam was produced by the water at  $\sim 67.3^\circ\text{C}$ . After the power was turned on (25 V) for 15 s, the fog was completely removed (as shown in Supporting Information). Furthermore, the temperature was measured under different voltages and a contour map of 30 V was presented, and showed in Fig. 6c and d. High voltage led to a fast rise of temperature, the defogging process started within 1 s after turn on the DC power

(30 V), and was completed in  $\sim 10\text{ s}$ . When the voltage was lower than 20 V, it required a very long time to reach the “starting” temperature, and failed for 10 V (fogging is faster than defogging). Due to the exclusion of the electrodes, the size of the contour map of temperature was  $\sim 1\text{ cm} \times 2\text{ cm}$ . Both of the edges of the contour map showed lower temperature. As mentioned in the experimental section, the edges of the sample were coated with silver paste to fabricate electrodes. The different radiative and convective heat loss from the different materials of surface might be the reason for lower temperature. After excluding the contiguous areas of electrodes, the surface temperature was  $50.8 \pm 1.5^\circ\text{C}$  in the central region. Most of the area exhibits a similar color, indicating a uniform temperature distribution. In general, the defogger we present had a fast defogging time under safety voltage (36 V), superior to that of other graphene-based defoggers demonstrated by previous reports [32,38,39]. (e.g., needing 21 s under 30 V). This was attributed to the direct synthesis and localized catalyzation. Due to the closer contact between graphene and quartz, defoggers fabricated by directly grown graphene had a better defogging response and robustness compared with transferred graphene [32,33]. Uniform and high coverage of graphene were ensured by localized catalyzation, which was the key for fast and uniform heating of the whole device. Consequently, these two main reasons led to better defogging behavior of our graphene-based defogger.

Moreover, the metal-assisted growth method always remains a technical challenge in removing the metal contamination completely. In this work, nanoparticles were introduced by spin-coating and slightly contact with quartz. Most of the nanoparticles gradually evaporated during the high temperature growth process above the melt point of Cu. Others could be partly removed by soaked the samples in alcohol without significantly affecting the electrical performance. The remains were extremely sparse, but still might be problems for devices. As for some applications such as heating device or transparent conducting electrodes, the effect of residual Cu could be negligible. This method still solved one of the main problem in this respect, which was the uncontrollability of the uniform catalysis when synthesized large-scale samples. Moreover, as the contact between Cu nanoparticles and quartz was weak, finding a more efficient method to clear them was



**Fig. 6.** (a) Schematically illustration of the single-layer graphene defogger; (b) Photographs of defogging process with input voltage of 25 V, corresponded video can be found in supporting information; (c) Temperature curve of the graphene glass (best experimental parameters) when input different voltages; (d) Contour map (9 points) of the saturation temperature on the surface of graphene glass while applying an input voltage of 30 V. (e) Defogging time under different input voltage, we measured 5 times under each voltage and calculated the average value, error bars refer to the variance of the data.

expectable.

#### 4. Conclusion

In summary, we have developed a Cu-nanoparticle-assist method (N-CVD) for the growth of large-scale and uniform graphene films on quartz substrates. With the assist of the “catalytic seeds”, the growth rate of graphene films was significantly improved and single-layered graphene film with low-defect was prepared after 240 min. Moreover, a detail growth model based on localized catalyzation was established. High quality graphene glass with low square resistance and high transmittance was successfully prepared using this method. Then We fabricated a defogger based on graphene glass, which can completely remove fog within 10 s when a safe voltage of 30 V was applied. Furthermore, given the benefit of localized catalyzation, this approach also opened up a new pathway to the growth of graphene on other dielectric substrates such as  $\text{SiO}_2$  and  $\text{Al}_2\text{O}_3$ . In essence, this work demonstrated an industrially feasible approach for the scalable growth of graphene films on quartz, further showing great potential of graphene glass in future applications.

#### CRediT authorship contribution statement

**Yue Li:** Conceptualization, Methodology, Validation, Formal analysis, Investigation, Writing - original draft, Visualization. **Yunbiao Zhao:** Investigation, Validation, Writing - review & editing. **Yi Chen:** Resources, Validation. **Yuhan Chen:** Validation. **Danqing Zhou:** Validation. **Ziqiang Zhao:** Supervision, Project administration, Funding acquisition, Writing - review & editing.

#### Declaration of Competing Interest

All authors have no conflicts of interest to declare.

#### Acknowledgement

This work was supported by National Natural Science and Foundation of China (NSFC) with numbers of 11875077.

#### Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.apsusc.2020.147082>.

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